



RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

Upgrading VCF 9.0.1 → 9.1 Procedure & Homelab Feasibility

The verified Broadcom 9.0.x→9.1 upgrade sequence and prerequisites, plus an honest feasibility assessment for a nested, brownfield, non-HCL homelab — where the precheck and the new VCF Management Services cluster are the real gates. No guessing.

v1.0

VMware Cloud Foundation 9.1

Runbook — Upgrading VMware Cloud Foundation 9.0.1 → 9.1 (with Homelab Feasibility Assessment)

Prepared by: Virtual Control LLC **Date:** July 27, 2026 **Source environment:** VMware Cloud Foundation 9.0.1 (nested, brownfield-imported, single-host homelab) **Target:** VCF 9.1

Read §2 (Feasibility) BEFORE committing. The upgrade **path** 9.0.x → 9.1 is supported, but this specific lab (nested, non-HCL, brownfield-imported, with the active cluster-store loop) faces **real precheck and resource blockers**. The **precheck + the VCF Upgrade Planning Tool are the empirical gate** — do those first; do **not** assume it will pass.

1. The Big Architectural Change in 9.1

VCF 9.1 **removes the standalone Fleet Management Appliance** (introduced in 9.0) and replaces it with **VCF Management Services** — a **new multi-node cluster** running **Fleet lifecycle + SDDC lifecycle + the software depot + the License Server**, deployed **during** the upgrade. The old Fleet Management appliance and the standalone VCF Identity Broker VMs are **decommissioned** as part of the process. (Source: Broadcom "Upgrading to VCF 9.1"; KB 440630.)

Consequence: the upgrade is not a simple in-place bump — it **stands up new infrastructure** (VCF Management Services + License Server) mid-flight. That is the heart of the feasibility question.

2. Feasibility for THIS Lab — the honest assessment

Supported in principle (9.0.1 is a valid 9.1 source). **But three grounded concerns:**

Concern	Why it bites this lab	Verified prereq it comes from
VCF Management Services footprint	New multi-node cluster needs 12–30 static IPs (forward and reverse DNS each) + a License Server (A + PTR). Its CPU/RAM is published only in the Planning Workbook — not guessed here — but a nested host already running vCenter/NSX/SDDC/Ops/Fleet/Aria has little headroom	"Plan 12–30 static IPs for VCF Management Services"; License Server mandatory
Non-HCL hardware	Prereq = "validate each host against the 9.1 HCL; confirm servers haven't dropped off." Nested ESXi is not on the HCL (the "Storage controller not on HCL" alert already fires). Stale/failed vSAN HCL is a documented precheck staller	"Validate each host model against the Broadcom compatibility guide for 9.1"

Concern	Why it bites this lab	Verified prereq it comes from
Clean-precheck gate on a degraded/brownfield env	Broadcom: "treat the precheck as a gate with no exceptions ... zero errors and no unresolved warnings." This env is brownfield-imported and has the active cluster-store DKVS loop — the very thing we hope 9.1 fixes may fail the precheck (chicken-and-egg)	"Run SDDC Manager precheck; must pass clean"

Verdict (no guessing): *possible on paper, but I would not expect a clean precheck without significant remediation, and the VCF Management Services deployment is a genuine resource risk. The definitive answer is empirical* — run \$5 Phase 0 (Planning Tool + precheck) and let the results decide. Broadcom's own maxim applies: "A VCF upgrade doesn't go sideways during the upgrade — it goes sideways in the week before."

3. Prerequisites (verified Broadcom list)

- **Backups:** fresh **SDDC Manager, vCenter, NSX** backups; confirm restore capability. (*We have recent vCenter + SDDC backups — take fresh ones at go-time.*)
- **Component minimums:** VCF Operations must be the upgrade-eligible 9.0.x build (from 9.0.1 it already is); SDDC Manager / vCenter / NSX / ESX on a valid 9.0.x → 9.1 path.
- **Networking / IPs:** **1 temporary IP per vCenter** (side-by-side upgrade) + **12–30 static IPs** for VCF Management Services; **every IP needs forward AND reverse DNS**. Internal service range default **198.18.0.0/15** (changeable to 240/250.0.0.0/15).
- **License Server:** mandatory new component; **needs an A record and a PTR** in DNS.
- **Certificates:** valid, correct FQDNs, **won't expire mid-window**; service/admin passwords valid throughout. (*Note: vCenter root expires Aug 10 — rotate before, per the credential items.*)
- **DNS/NTP:** DNS resolves all FQDNs **bidirectionally**; NTP operational. (*NTP hardened 2026-07-27; DNS has a history of dnsmasq/AAAA issues here — verify carefully.*)
- **HCL:** validate hosts against the 9.1 compatibility guide; **refresh the vSAN HCL database** (manual/offline for air-gapped).
- **Cloud Proxy:** operational for VCF Operations integration.
- **Depot:** stage the **9.1 bundles** in the (offline) depot; confirm the **download entitlement** covers 9.1.
- **Planning & Preparation Workbook:** complete the network/addressing/hardware rows.

Common blockers (documented): certs expiring mid-window · missing **reverse DNS** (blocks VCF Management Services) · stale vSAN HCL · no temporary vCenter IP.

4. The Upgrade Sequence (verified — Broadcom 9.0.x → 9.1 order)

(Skip components not present in this environment.)

1. VCF Identity Broker (redploy if needed)
2. **VCF Operations & Cloud Proxy** ← the sequence starts here 3–6. vSphere Replication · Live Site Recovery · vSAN Data Protection · Avi LB (*skip if absent*)
3. **SDDC Manager → 9.1, then DEPLOY VCF Management Services + License Server** ← pivotal step
4. VCF Identity Broker → 9.1
5. VCF Automation

6. VCF Operations Orchestrator 11–13. VCF Ops for Networks · for Logs · HCX 14–15. NSX Global Manager · **NSX Manager**
7. **vCenter** 17–18. vSphere Kubernetes Service · Supervisor (*if present*)
8. vSAN Witness Host (*if present*)
9. **ESX**
10. **NSX Edge & NSX Finalize** 22–24. VMware Tools · VM Compatibility · vSAN On-Disk Format / File Service

After VCF Operations reaches 9.1 you cannot manage certs/passwords for VCF Automation, Identity Broker, Ops-for-logs, Ops-for-networks until those are also on 9.1 — so do credential work before starting.

5. The Plan (phased — gate before you commit)

Phase 0 — Feasibility gate (*do this FIRST; non-destructive*)

1. Run the [VCF Upgrade Planning Tool](#) against this environment for a tailored path + blocker list.
2. Pull **VCF Management Services sizing** from the **9.1 Planning & Preparation Workbook**; compare vCPU/RAM/IP needs to **actual free capacity** on the nested hosts. **If it doesn't fit, stop here.**
3. Confirm **9.1 download entitlement** and stage bundles in the offline depot.
4. **Run the SDDC Manager 9.1 precheck.** This is the empirical answer. Record every error/warning. **If it can't reach zero errors / no unresolved warnings, the upgrade is not feasible as-is** — and some blockers (nested HCL, the DKVS loop) may not be cleanly resolvable.

Phase 1 — Preparation (*only if Phase 0 is clean*)

- Fresh backups (SDDC, vCenter, NSX) + verified restore path.
- DNS: create the 12–30 forward+reverse records + the License Server A/PTR.
- Rotate any credential/cert that would expire in the window (vCenter root, etc.) **before** starting.
- Refresh vSAN HCL DB; confirm NTP/DNS bidirectional health.
- Complete the Planning & Preparation Workbook.

Phase 2 — Execute

- Work the §4 sequence via the lifecycle workflow: VCF Operations → SDDC Manager (+ deploy VCF Management Services + License Server) → NSX → vCenter → ESX → Edge → finalize.
- Monitor each task to Successful; use the workflow's retry on a failed task before restarting.

Phase 3 — Post

- Verify all components at 9.1; the old Fleet Mgmt + Identity Broker VMs are decommissioned automatically.
- **Confirm the cluster-store (DKVS) loop is resolved** (the primary reason for this upgrade) — `clusterAdmin c1uster status` + the loop monitor should quiesce.

6. Sources (Broadcom)

- [Upgrading to VMware Cloud Foundation 9.1 \(techdocs\)](#)
- [KB 440630 — Upgrade Sequence and Related Issues for VCF/vSphere Foundation 9.1](#)
- [VCF Upgrade Planning Tool](#)

- VCF 9.1 Planning & Preparation Workbook (referenced by the upgrade docs — the authoritative source for VCF Management Services sizing)

What is deliberately NOT stated in this runbook (would be a guess): the exact **VCF Management Services CPU/RAM/node sizing** (Broadcom publishes it only in the Planning Workbook — get it there), and any assertion that this lab's precheck **will** pass. Both are resolved empirically in Phase 0.

Document Information

Field	Value
Document ID	VC-RUNBOOK-VCF-91-UPGRADE-20260727
Version	1.0
Issued	July 27, 2026
Author	Virtual Control LLC
Classification	Internal
Platform	VMware Cloud Foundation 9.0.1 -> 9.1

Revision History

Version	Date	Notes
1.0	2026-07-27	Initial runbook built from the verified Broadcom 9.1 upgrade docs (techdocs "Upgrading to VCF 9.1", KB 440630): the 24-step sequence, prerequisites, and the new VCF Management Services deployment (replacing the Fleet appliance; 12-30 IPs + License Server). Includes an honest homelab feasibility assessment (nested/non-HCL/brownfield/DKVS-loop concerns) and a phased plan gated on the Planning Tool + SDDC precheck. VCF Management Services CPU/RAM sizing deliberately NOT stated (Planning Workbook only) - no guessing.

Document End